SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.2: Mechanical Properties of Materials — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.2: Mechanical Properties of Materials
Driving Question	DRIVING QUESTION: Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon: Why Does a Rubber Band Snap Back But a Plastic Bag Stays Stretched?**	Students observed everyday Kenyan materials — rubber bands, polythene bags, metal wire, sisal string, and a steel spring — being stretched and released. They noted which materials returned to their original shape and which remained deformed or broke. DQB was introduced: students posted initial questions and predictions on the class Driving Questions Board. Teachers should note which misconceptions surface here (e.g., 'heavier materials are always stronger') as these will need revisiting in later lessons. Students also began their first Anchor Model sketch showing what they think happens inside a material when it is pulled.	Students learned that different materials respond differently to the same applied force — some recover fully (like the rubber band), some deform permanently (like the polythene bag), and some fracture (like overstretched sisal). This lesson establishes the anchoring phenomenon for the entire unit. Key vocabulary introduced: force, deformation, recovery. Teachers should emphasise that the 'why' behind these observations is the central puzzle of the unit and will not be fully resolved until Lesson 6.	Students used their initial observations to begin explaining the anchoring phenomenon in their own words: 'The rubber band goes back because…' Crucially, these early explanations will be incomplete and even incorrect — that is intentional. Teachers should affirm curiosity and effort rather than correctness at this stage. Pedagogical note: Cold-call at least 4 students to share predictions; record key ideas on the DQB without judgment. Allow 30 seconds of wait time after each question. Connect every student response back to the driving question: 'So are you saying the material's inside structure matters?'
Lesson 2 **Stretching the Truth: Elastic vs. Plastic Behaviour in Real Materials**	Students stretched and released a range of materials — rubber bands (elastic behaviour), modelling clay or ugali dough (plastic behaviour), copper wire (elastoplastic behaviour) — and recorded whether each returned to its original shape. They plotted simple force-vs-extension sketches by hand. DQB update: students added new questions such as 'Is there a	Students learned the precise definitions of elasticity (a material returns to its original shape after the deforming force is removed) and plasticity (a material permanently deforms and does not recover). They learned that the elastic limit is the boundary between these two behaviours for any given material — beyond it, elastic materials begin to behave plastically. Kenyan context:	Students explained why a rubber band returns to shape (elastic behaviour — internal bonds restore to original configuration) while ugali dough stays deformed (plastic behaviour — bonds are permanently rearranged). Teachers should use this lesson to directly address the first half of the driving question: 'stretch and snap back vs. permanently deform.'

	point where even a rubber band won't come back?' Teachers should circulate and check that students are distinguishing between the deformation being temporary vs. permanent, not just 'big vs. small stretch.'	builders choosing steel rebar vs. timber for a Nairobi apartment block need to understand which material will deform elastically under load and which will yield permanently.	Pedagogical note: Ask students to rank the tested materials from most elastic to most plastic and justify their ranking — this surfaces deeper conceptual reasoning beyond surface-level observation.
Lesson 3 **Forces in Balance: Tension, Compression, and How Structures in Kenya Are Loaded**	Students observed and manipulated model structures (using straws, string, and cardboard) representing common Kenyan construction scenarios — a roof truss, a suspension footbridge over a small river, a market stall frame. They identified which members were in tension (being pulled apart) and which were in compression (being pushed together). DQB update: students revised earlier questions and added new ones: 'Does it matter whether a material is being pulled or pushed?' Model revision: students updated their Anchor Model to show arrows indicating tension vs. compression forces.	Students learned that structural materials in real Kenyan buildings and bridges experience two primary types of loading: tension (stretching force — e.g., cables in the Likoni Ferry suspension system) and compression (squeezing force — e.g., concrete columns in Nairobi CBD buildings). Different materials are best suited for different loading types: steel cables excel under tension; concrete excels under compression but is weak in tension. This lesson connects material properties to real engineering decisions.	Students explained why engineers use both steel and concrete together in reinforced concrete (used widely in Kenyan construction) — the steel handles tension while the concrete handles compression. Teachers should link this back to the driving question: 'choosing the right material for a safe structure.' Pedagogical note: Cold-call at least 3 students to point out tension vs. compression members in a displayed photograph of a Kenyan bridge or building. Allow 20 seconds of wait time. If students confuse tension and compression, return to the physical model and have them feel the difference with their hands.
Lesson 4 **Hooke's Law — Measuring Spring Constant and Finding the Elastic Limit**	Students conducted a structured investigation: they hung known masses (50 g, 100 g, 150 g… up to 500 g) from a spring and measured the extension at each step using a ruler. They plotted Force (N) vs. Extension (m) graphs and identified (1) the linear region where Hooke's Law holds ($F = kx$), (2) the elastic limit — the point beyond which the graph curves — and (3) the permanent deformation region. DQB update: students answered the question 'Is there a point where even a rubber band won't come back?' using their graph evidence. Model revision: students added the elastic limit threshold to their Anchor Model.	Students learned that (1) for a spring loaded within its elastic limit, extension is directly proportional to applied force — this is Hooke's Law: $F = kx$, where k is the spring constant (N/m); (2) the spring constant k is the gradient of the linear portion of the F-vs-x graph and measures stiffness — a higher k means a stiffer spring; (3) beyond the elastic limit, the material no longer obeys Hooke's Law and permanent deformation begins. Teachers should ensure students can calculate k from their graph gradient, not just recall the formula.	Students explained the shape of their Force-vs-Extension graph by linking each region to a physical behaviour: linear region = elastic (Hooke's Law obeyed), curve onset = elastic limit reached, plateau/break = plastic deformation or fracture. Kenyan context: mattress spring manufacturers in Nairobi's Industrial Area select springs with specific k values to support different body weights without exceeding the elastic limit. Pedagogical note: Common error — students confuse 'extension' (change in length) with 'total length.' Check graphs carefully. Ask: 'What does the gradient of your graph tell you about the spring?' Cold-call 3 students.
Lesson 5 **Stress, Strain, and Young's Modulus: Predicting Which Rope Will Snap**	Students compared two ropes of different cross-sectional areas (e.g., thin sisal twine vs. thick nylon rope used by Kenyan fishermen at Lake Victoria) under increasing loads. They measured force, original length, cross-sectional area, and extension to calculate stress ($\sigma = F/A$, in Pascals) and strain ($\varepsilon = \Delta L/L_0$, dimensionless) for each. They observed	Students learned that stress ($\sigma = F/A$) describes the internal force per unit cross-sectional area — it tells us how 'hard' a material is being pulled per unit area, not just the total force. Strain ($\varepsilon = \Delta L/L_0$) describes the fractional change in length — a dimensionless ratio. Young's Modulus ($E = \sigma/\varepsilon$, in Pascals) is the ratio of stress to strain in the elastic region and is a material	Students explained why a thin rope snaps before a thick rope under the same force: same force over a smaller area produces higher stress, which more quickly exceeds the material's ultimate tensile stress. They used Young's Modulus to compare the stiffness of different materials quantitatively. Pedagogical note: Stress and strain are consistently among the hardest concepts in

	that the thinner rope reached a higher stress value for the same applied force and snapped first. DQB update: students answered 'Does the thickness of a material matter for how strong it is?' with quantitative evidence.	constant that measures stiffness independent of the object's dimensions. A higher E means a stiffer material. Kenyan context: structural engineers calculating whether steel rebar in a Rift Valley bridge can handle earthquake-induced stress use E for steel (≈200 GPa).	this unit — students often confuse stress with force and strain with extension. Require students to write the units of each quantity during every calculation. Cold-call 4 students to explain in their own words what 'stress' means physically, allowing 30 seconds of wait time each.
Lesson 6 **Final Explanation: Elastic Potential Energy, Elastic Limits, and Choosing the Right Material for a Safe Structure**	Students revisited the anchoring phenomenon materials from Lesson 1 (rubber band, polythene bag, steel spring, sisal string) and applied all unit concepts to explain each material's behaviour fully. They calculated elastic potential energy stored in stretched springs using $E_p = \frac{1}{2}kx^2$, compared values for materials above and below the elastic limit, and used stress, strain, and Young's Modulus data to recommend materials for a given Kenyan engineering scenario (e.g., cables for a footbridge in Kisumu, roofing frame for a community hall in Kericho). Final DQB review: students marked which driving questions had been answered and identified any remaining questions. Final Model revision: students produced a complete, annotated Anchor Model showing elastic behaviour, plastic behaviour, Hooke's Law region, elastic limit, stress, strain, and energy storage.	Students learned that elastic potential energy stored in a stretched spring is $E_p = \frac{1}{2}kx^2$; that the elastic limit is the maximum extension a material can undergo and still return to its original shape; and that material selection for safe Kenyan structures requires evaluating multiple properties together — stiffness (k or E), strength (ultimate tensile stress), elastic vs. plastic behaviour, and energy storage capacity. They also learned that exceeding the elastic limit in a structural material is a failure condition that engineers must design against. This lesson directly and fully answers the unit driving question.	Students produced a final written or verbal explanation of the anchoring phenomenon: 'A rubber band stretches and snaps back because it is elastic — it stores energy as $E_p = \frac{1}{2}kx^2$ and releases it fully when the force is removed, as long as the elastic limit is not exceeded. A plastic bag deforms permanently because the applied stress causes the material to behave plastically beyond its elastic limit. Choosing the right material for a safe structure in Kenya means selecting a material whose elastic limit, Young's Modulus, and energy capacity match the expected loads.' Pedagogical note: Use this lesson's final explanations as a formative assessment. Compare them to students' Lesson 1 predictions on the DQB to make the learning journey visible. Celebrate growth in scientific reasoning, not just correct answers.